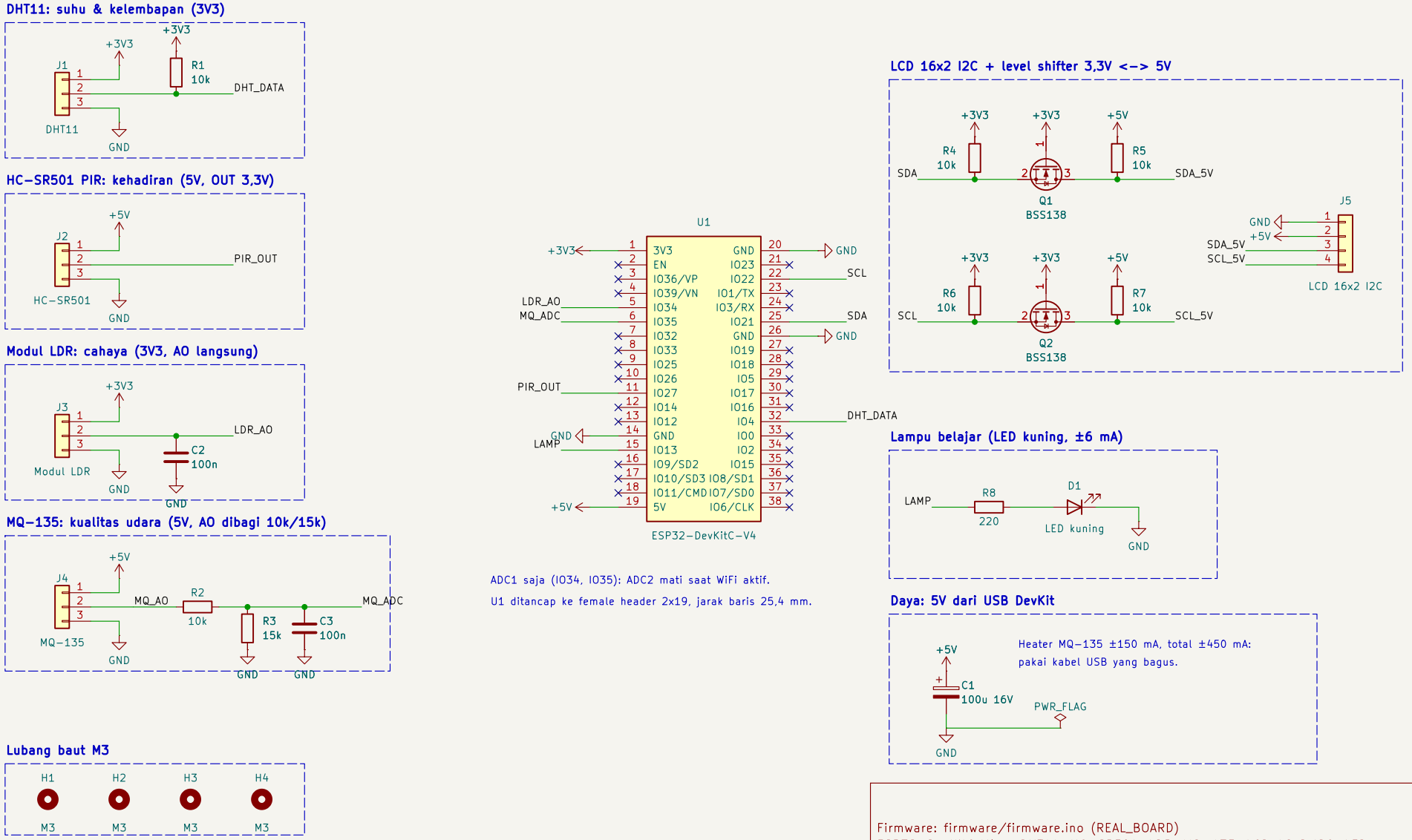

DHT11: suhu & kelembapan (3V3)
+3V3
R1
10k
DHT_DATA
J1
DHT11
GND
HC-SR501 PIR: kehadiran (5V, OUT 3,3V)
+5V
J2
PIR_OUT
HC-SR501
Modul LDR: cahaya (3V3, AO langsung)
J3
LDR_AO
C2
100n
Modul LDR
MQ-135: kualitas udara (5V, AO dibagi 10k/15k)
J4
MQ_AO
R2
10k
R3
15k
C3
100n
MQ_ADC
MQ-135
Lubang baut M3
H1
H2
H3
H4
M3
U1
3V3
EN
IO36/VP
IO39/VN
IO34
IO35
IO32
IO33
IO25
IO26
IO27
IO14
IO12
GND
IO13
IO9/SD2
IO10/SD3
IO11/CMD
5V
IO23
IO22
IO1/TX
IO3/RX
IO21
IO19
IO18
IO5
IO17
IO16
IO4
IO0
IO2
IO15
IO8/SD1
IO7/SD0
IO6/CLK
LDR_AO
MQ_ADC
PIR_OUT
LAMP
SCL
SDA
DHT_DATA
ESP32-DevKitC-V4
ADC1 saja (IO34, IO35): ADC2 mati saat WiFi aktif.
U1 ditancap ke female header 2x19, jarak baris 25,4 mm.
LCD 16x2 I2C + level shifter 3,3V <-> 5V
R4
10k
R5
10k
SDA
SDA_5V
Q1
BSS138
R6
10k
R7
10k
SCL
SCL_5V
Q2
BSS138
J5
LCD 16x2 I2C
Lampu belajar (LED kuning, ±6 mA)
R8
220
D1
LED kuning
Daya: 5V dari USB DevKit
C1
100u 16V
PWR_FLAG
Heater MQ-135 ±150 mA, total ±450 mA:
pakai kabel USB yang bagus.
Firmware: firmware/firmware.ino (REAL_BOARD)
ESP32-DevKitC V4 + DHT11, HC-SR501, LDR, MQ-135, LCD 16x2 I2C, LED
UTS/UAS IoT
Sheet: /
File: studysphere.kicad_sch
Title: StudySphere Room Node
Size: A4
Date: 2026-10-20
Rev: 1.0
KiCad E.D.A. 10.0.7
Id: 1/1